

Crash Course in JavaScript

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Description:

Crash Course in JavaScript is a fast-paced, beginner-friendly guide designed to help you quickly grasp the fundamentals of JavaScript. It introduces essential concepts with simple explanations and practical examples so you can start building real projects quickly.

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1. INTRODUCTION

JavaScript is one of the most important programming languages in modern web development. It allows developers to create interactive and dynamic websites. It works in browsers and also on servers using Node.js.

2. GETTING STARTED

To start with JavaScript, you need:

- A browser (Chrome, Firefox)
- A code editor (VS Code)

First code:

```
console.log("Hello, JavaScript!");
```

3. VARIABLES AND DATA TYPES

Variables store data.

Examples:

```
let name = "John";
```

```
const age = 25;
```

```
var city = "Dakar";
```

Data types:

- String
- Number

- Boolean
- Array
- Object
- Null / Undefined

4. OPERATORS

Used for calculations and logic.

```
let a = 10;
```

```
let b = 5;
```

```
a + b
```

```
a > b
```

```
true && false
```

Types:

- Arithmetic
- Comparison
- Logical

5. CONTROL FLOW

Used to make decisions.

```
if (age >= 18) {  
  console.log("Adult");  
} else {  
  console.log("Minor");  
}
```

Loop:

```
for (let i = 0; i < 5; i++) {  
  console.log(i);  
}
```

6. FUNCTIONS

Reusable blocks of code.

```
function greet(name) {  
  return "Hello " + name;  
}  
const add = (a, b) => a + b;
```

7. ARRAYS AND OBJECTS

Array:

```
let fruits = ["Apple", "Banana", "Mango"];
```

Object:

```
let user = {  
  name: "John",  
  age: 30  
};
```

8. DOM MANIPULATION

Used to interact with HTML.

```
document.getElementById("title").innerText = "Hello World";
```

9. EVENTS

Used to handle user actions.

```
document.addEventListener("click", function () {  
  alert("Clicked!");  
});
```

10. ASYNCHRONOUS JAVASCRIPT

Used for API calls and delays.

Promise:

```
new Promise((resolve) => resolve("Success"));
```

Async/Await:

```
async function fetchData() {  
  let res = await fetch("https://api.example.com");  
}
```

11. ERROR HANDLING

```
try {  
  JSON.parse("invalid");  
} catch (error) {  
  console.log("Error detected");  
}
```

12. MINI PROJECT: TO-DO LIST

```
let tasks = [];  
function addTask(task) {  
  tasks.push(task);  
}
```

You can extend it with delete, update, and display features.

13. CONCLUSION

JavaScript is a powerful language for web development.

Mastering the basics allows you to move toward frameworks like React, Node.js, and modern full-stack development.

Practice is the key to mastery.